College Basketball In-game Metrics and Win Percentage

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Abstract

This report examines NCAA Division I men's basketball data to identify which in-game statistics most strongly impact team success, measured by win percentage. Using multiple linear regression and an AIC-based best subsets selection process, we developed a model with an adjusted R^2 of 0.82. Key predictors included defensive effective field goal percentage and turnover rate. While the model performs well, it occasionally predicts win percentages above 1 for elite teams, reflecting a limitation of linear regression on bounded data. BYU's 2025 season showed a predicted win percentage of 77%, close to their actual 72%, demonstrating relatively strong predictive performance. The results offer practical insights for coaching strategy and player recruitment.

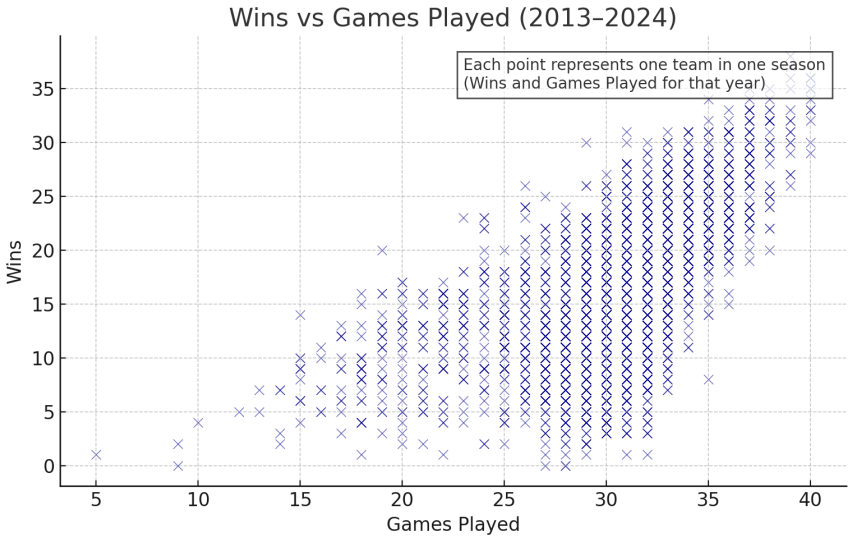

Figure 1: Scatterplot of Wins vs Games Played

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1 Introduction

College basketball, with its dynamic roster changes and competitive parity, presents a fascinating challenge for predictive modeling. The percentage of games won over a season is an important indicator often used to evaluate coaching effectiveness, predict tournament outcomes, and inform sports betting.

This study seeks to identify which in-game performance metrics—such as offensive and defensive efficiency, shooting percentages, and rebounding—have a significant impact on games won. Additionally, we aim to determine which metrics can be used to predict the win percentage of a Division I men's college basketball team. Using data from the 2013 through 2024 seasons, we aim to develop a model that predicts a team's success based on performance metrics from past seasons.

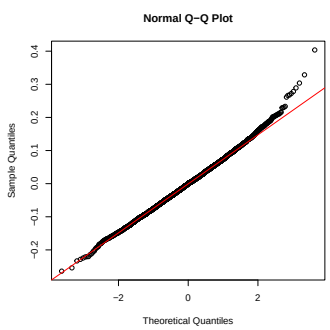
1.1 Data Description

This study used data from a College Basketball Dataset compiled by Andrew Sundberg, encompassing NCAA Division I men's basketball seasons from 2013 through 2025. The primary data file, cbb.csv, includes seasons from 2013 to 2019 and from 2021 to 2024, while cbb25.csv contains data for the 2025 season as of March 18, 2025. The 2020 season was excluded due to the COVID-19 pandemic.

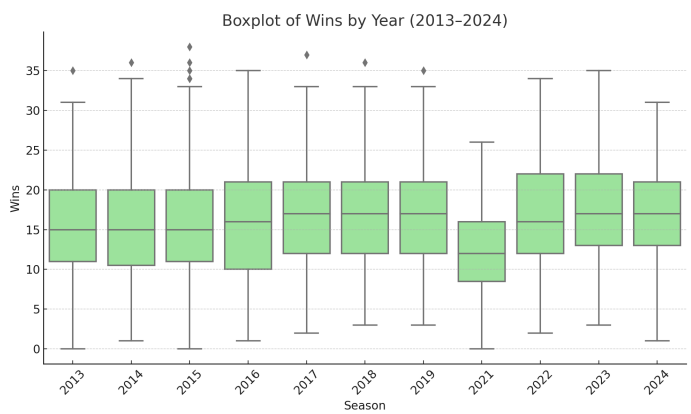
The initial dataset included 24 variables. Of these, 17 were relevant to our analysis objectives. After model selection, we narrowed the model down to 9 predictors and filtered the data to retain 12 total variables for the final analysis.

Table 1: ***Data Table: Variable Descriptions***

#	Feature	Description
1	Team	The Division I college basketball school
2	G	Number of games played
3	W	Number of games won
4	Win_Pct	Win Percentage (calculated as W/G)
5	ADJOE	Adjusted Offensive Efficiency (an estimate of the offensive efficiency, measured in points scored per 100 possessions)
6	ADJDE	Adjusted Defensive Efficiency (an estimate of the defensive efficiency, measured in points allowed per 100 possessions)
7	EFG_O	Effective Field Goal Percentage Shot
8	EFG_D	Defensive Effective Field Goal Percentage
9	X2P_O	Two-Point Shooting Percentage
10	X2P_D	Two-Point Shooting Percentage Allowed
11	X3P_O	Three-Point Shooting Percentage
12	X3P_D	Three-Point Shooting Percentage Allowed
13	ADJ_T	Adjusted Tempo (an estimate of the tempo in possessions per 40 minutes a team would have against the average opponent)
14	TOR	Turnover Rate
15	TORD	Steal Rate
16	ORB	Offensive Rebound Rate
17	DRB	Offensive Rebound Allowed Rate



(a) Q-Q Plot of Model Residuals



(b) Boxplots of Win Count by Season

Figure 2: Normality and Distribution Diagnostics

Looking at the distribution of wins by year, we can see that the distribution appears to be relatively normal. Further analysis of a QQ-plot for normality shows that the normality assumption is in tact despite some upper-tail skewness. Other assumptions of multiple linear regression, including linearity, independence, and equal variance were met after further analysis.

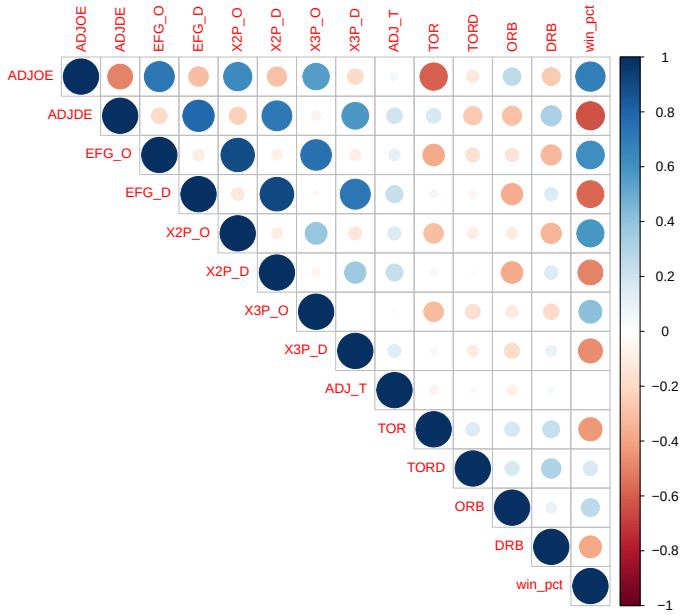


Figure 3: Correlation plot of potential variables

Table 2: Variance Inflation Factors (VIF) for Model Predictors

Predictor	**VIF**
Defensive Efficiency	4.411
Offensive Effective Field Goal %	1.304
Defensive Effective Field Goal %	7.636
2-Point Field Goal % Allowed	5.692
Adjusted Tempo	1.090
Turnover Rate	1.260
Steal Rate	1.641
Offensive Rebound Rate	1.270
Offensive Rebounds Allowed Rate	1.624

To assess potential issues with multicollinearity, we examined the correlation structure of the selected predictors and calculated their variance inflation factors (VIFs). While some predictors showed moderate VIF values, none exceeded the common threshold of 10, suggesting that multicollinearity is not a significant concern. Additionally, model selection via cross-validation reinforced the stability of the final model. Based on these diagnostics, we conclude that the multicollinearity assumption has not been violated.

2 Model Selection and Validation

Following several variable selection methods and cross-validation for out-of-sample performance, we determined the best model for analysis to be the following:
(1).

$$\begin{aligned}\text{Win Percentage}_i = \beta_0 &+ \beta_1 \times \text{Defensive Efficency}_i \\ &+ \beta_2 \times \text{Offensive Effective FG}_i \\ &+ \beta_3 \times \text{Defensive Effective FG}_i \\ &+ \beta_4 \times \text{2 Point FG Allowed}_i \\ &+ \beta_5 \times \text{Adjusted Tempo}_i \\ &+ \beta_6 \times \text{Turnover Rate}_i \\ &+ \beta_7 \times \text{Steal Rate}_i \\ &+ \beta_8 \times \text{Offensive Rebound Rate}_i \\ &+ \beta_9 \times \text{Offensive Rebound Allowed Rate}_i \ + \epsilon_i,\end{aligned} \tag{1}$$

where

$$\epsilon_i \sim N(0, \sigma^2).$$

We looked at several metrics and statistics to determine whether this would be the best model to predict the percentage of wins. This included adjusted R^2 and PSME. We used the following three model selection methods: AIC best subsets, BIC best subsets, and LASSO regression. In order to determine which of these models would work best, we ran a cross-validation for all three models and computed the predicted mean squared error for each. These values can be seen below:

Out-of-Sample PMSE Summary

Model	**Average PMSE**	**Standard Deviation**
OLS	0.008197	0.000455
AIC Best Subset	0.008193	0.000456
BIC Best Subset	0.008197	0.000459
LASSO	0.008388	0.000480

Looking at these results, we concluded that the model selected using AIC best subsets would be the best fit for our data. It contained the lowest predicted mean squared error, and the standard deviation of that error was also relatively low, indicating that the model performed reliably across different train-test splits. The model was also simpler than the ordinary least squares model, as it included fewer predictors, making it more interpretable and less prone to overfitting. Notably, it recorded the highest adjusted R^2 value at 0.82, suggesting that it explained a strong proportion of the variability in our response variable while penalizing for complexity.

These results suggest that the AIC-selected model performs, while remaining relatively simple. It has a strong predictive ability with fewer predictors, which is especially valuable when considering practical applications or future data collection. Additionally, the consistency of its performance across iterations increases our confidence in the out-of-sample application.

3 Analyses, Results, and Interpretation

The table below presents the summary of the model selected through the AIC-based best subsets approach:

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	0.1961	0.0511	3.84	0.0001
Defensive Efficiency	0.0020	0.0004	4.79	0.0000
Offensive Effective FG	0.0283	0.0005	59.51	0.0000
Defensive Effective FG	-0.0351	0.0012	-28.65	0.0000
2 Point FG Allowed	0.0069	0.0009	7.43	0.0000
Adjusted Tempo	0.0031	0.0004	7.26	0.0000
Turnover Rate	-0.0230	0.0007	-33.64	0.0000
Steal Rate	0.0247	0.0007	33.59	0.0000
Offensive Rebound Rate	0.0095	0.0003	27.60	0.0000
Offensive Rebound Allowed Rate	-0.0134	0.0005	-26.58	0.0000

Looking at this summary output, we can conclude that each predictor is statistically significant at the 0.05 significance level. The intercept has a value of 0.1961, which represents the predicted win percentage for a team with zero values across all nine predictors. Although such a scenario is not realistic in practice, the intercept provides a baseline reference point for the model and helps anchor the interpretation of the other coefficients. Among the predictors, defensive effective field goal percentage has the largest absolute beta coefficient. This means that for every one-unit increase in a team's field goal percentage allowed on defense, holding all other variables constant, the expected win percentage decreases by roughly 3.5%. This highlights the importance of strong defensive performance in contributing to overall team success.

A 95 percent confidence interval of the beta coefficients can be seen below:

0.95 Confidence Interval of Coefficients	2.5 %	97.5 %
(Intercept)	0.0960	0.2963
Defensive Efficiency	0.0012	0.0028
Offensive Effective FG	0.0274	0.0293
Defensive Effective FG	-0.0375	-0.0327
2 Point FG Allowed	0.0051	0.0087
Adjusted Tempo	0.0023	0.0040
Turnover Rate	-0.0244	-0.0217
Steal Rate	0.0232	0.0261
Offensive Rebound Rate	0.0088	0.0101
Offensive Rebound Allowed Rate	-0.0144	-0.0125

Examining the results from the 95% confidence intervals for the beta coefficients in our model, we observe that none of the intervals contain zero. This indicates that each predictor in the model is statistically significant and directly impacts win percentage. Notably, steal rate has a substantial influence: holding all other variables constant, we can be 95% confident that a one-unit increase in a team's steal rate would lead to an increase in their win percentage between 2.3% and 2.6%.

Using this model, we predicted the 2025 BYU men's basketball win percentage (note: 2025 data was not included in the original dataset used in cross-validation). Based on

BYU's observed values for the predictive variables, the model estimated a win percentage of 77%, with a 95% prediction interval ranging from 62% to 92%. This means that, accounting for model uncertainty and natural variation, we are 95% confident that BYU's actual win percentage would fall within that range. The model's fitted prediction of 77% was relatively close to BYU's actual win percentage of 72%, suggesting that the model performed reasonably well when applied to new data.

It is important to note that while this model can be used to predict win percentage, it may encounter issues when applied to extremely high-performing teams. This is because win percentage is naturally bounded between 0 and 1, but linear regression does not account for these limits. For example, Duke performed exceptionally well in the 2025 season, and our MLR model might overestimate its performance to the point of predicting a win percentage greater than 1 — a value that is not possible in reality.

This model more specifically useful for identifying which in-game statistics have the greatest impact on a team's success. Rather than serving as a perfect prediction tool, it offers valuable insight into the key areas that contribute to winning. College basketball programs could use a model like this to evaluate which aspects of performance they should prioritize in the upcoming season. For example, a coach might notice that offensive rebounds allowed have a strong negative effect on win percentage. This could lead them to focus on recruiting rebounding guards who consistently box out, or pursuing a dominant center to collect more rebounds.

4 Conclusions

Following initial analysis of our data, we determined multiple linear regression to be the best method for modeling win percentage and identifying the importance of various in-game statistics in predicting success for Division I men's college basketball teams. Using AIC best subsets and cross-validation, we selected an optimal model containing nine predictors: adjusted defensive efficiency, offensive effective field goal percentage, defensive effective field goal percentage, two-point shooting percentage allowed, adjusted tempo, turnover rate, steal rate, offensive rebound rate, and offensive rebounds allowed rate. A college program looking to improve its win percentage might use this model to prioritize certain metrics in recruiting and daily practice.

The beta coefficients were mostly consistent with our expectations. However, one exception was the positive coefficient for two-point field goal percentage allowed, where we initially expected a negative relationship. We suspect this result may be due to its correlation with offensive effective field goal percentage, which carries a strong positive coefficient in the model. This overlap in predictive power could have caused the model to assign a counterintuitive positive sign to the two-point percentage variable.

We found that the model performed relatively well in terms of out-of-sample prediction, with an adjusted R^2 of 0.82 and a predictive mean squared error of 0.008193. One limitation of the model is that it relies on in-game statistics, which are not available before a season begins. However, programs can still use past data to estimate how improving one or more key variables might affect future performance.

Another limitation is that win percentage is bounded between 0 and 1, while linear regression does not account for these bounds. As a result, teams with extreme predictor values might receive unrealistic predictions above 1 or below 0. However, such extreme predictions are unlikely for most observations. Beta-binomial regression may address this, which we would like to explore further.

APPENDIX

We used generative AI in order to receive critical feedback on our report, and to receive input on how we could improve our writing.

- Prompts include:
  - What are the strengths and weaknesses of our report?
  - What are potential better methods of regression modeling?
- Recommendations included:
  - Slight inconsistencies in variable naming (e.g., "effective offensive field goal percentage allowed" vs. "defensive effective field goal percentage").
  - You could be slightly more cautious when interpreting the intercept
  - Visuals are informative but could benefit from more detailed captions (e.g., Figure 3 doesn't explain what to look for)
- Rational for dismissal:
  - We felt that we had addressed the main idea or interpretations of important figures and tables. For example, on the same page as Figure 3 we mention multicollinearity and discuss conclusions drawn from VIF results.
- Incorporated adjustments:
  - We went through to ensure variable names were consistent.
  - We also corrected any grammar errors or awkward sentence flow.
  - Additionally, it was suggested that a beta-binomial regression could be a better solution to model this. We mentioned this in our report.
  - Lastly, we clarified the intercept interpretation.

Additional Diagnostic Plots:

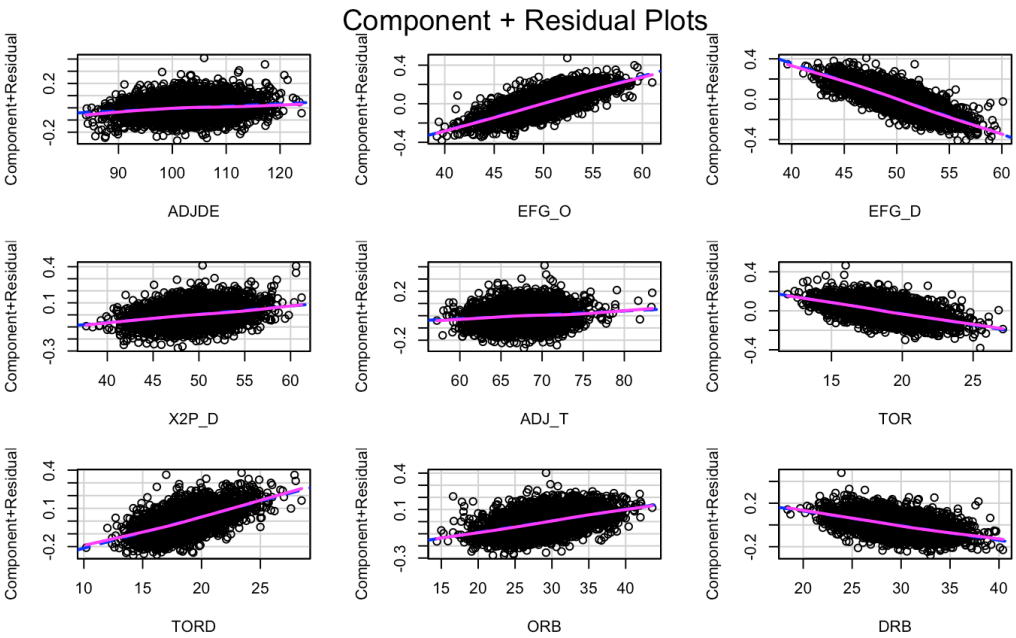
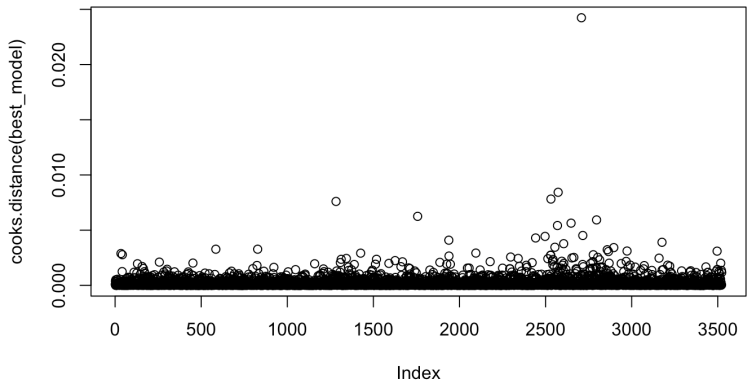


Figure 4: Component + Residual Plots



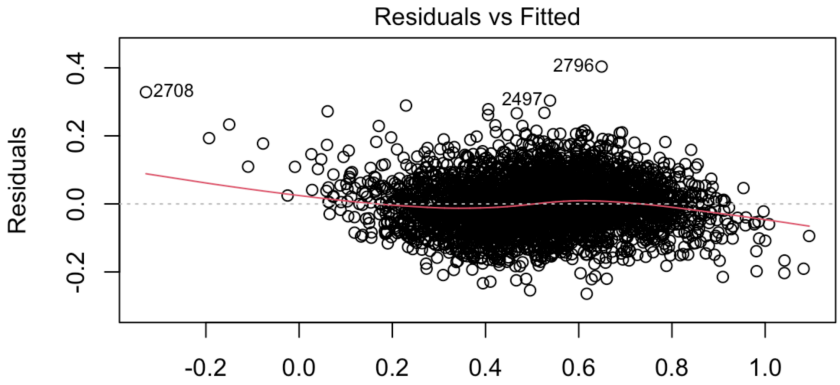


Figure 5: Cook's Distance (top) and Residuals vs. Fitted (bottom)